

Financial Data Pipeline · Ten Cuts, Measured

From public filings to a cross-company comparison table where **every cell states its own trustworthiness** — every step reproducible · 2026-09 · github.com/laa1991/fin-data-rulers

Problem

- Errors in an investment-data pipeline **do not raise alarms**: wrong source, wrong basis, wrong point-in-time, wrong extraction all run fine, look right — **and the conclusion is wrong**.
- So every cut ships with a **machine-decidable check** instead of a human judge — and each failure mode is **deliberately injected once** to see whether the check catches it.

What was built (ten cuts, four groups)

Rulers ①–④ source reconciliation · adjustment basis · reporting period vs announcement date · scale-out to $n=50$
Applications ⑤–⑦ citation-traceable RAG · automated report with number-to-source checking · **point-in-time (as-of) discipline**

Construction ⑧–⑩ statement tables reconstructed with **accounting-identity self-verification** · cross-industry **concept disambiguation** · the comparison table
Engineering multi-source + backoff + cache + resumable fetch · one command builds the table · whole chain reproducible

Hardest readings

- Source**: two independent sources reconciled day by day, **242/242 trading days**, max absolute close difference **0.000000**.
- Basis**: annual return gap **3.32 pts**; **28/50** names flip momentum sign; extreme case **-9.63% ↔ +28.32%** (the basis decides whether it is a loss or a gain).
- Point-in-time**: filings are announced on average **30.8 trading days** after period end; the union of four windows covers **49.6%** of trading days.
- Q&A**: citation present **1.000** / citation correct **1.000** / refusal **1.000**; automated-report number traceability **0.804 → 1.000** (zero fabrication).
- As-of discipline**: the unconstrained arm violates the cutoff in **36%** of answers and is wrong on **12/12** "not yet disclosed" questions; **0.000** once enforced.
- Table reconstruction**: **228** identity checks; **all 40 failures traced back to my extraction** — none to the statements themselves. The check tells me where not to trust, rather than judging for me.
- Concept disambiguation**: across three issuers the same line item is labelled differently in **11 concepts** — units differ too (CNY vs CNY millions); cross-company identities pass **27/27**; **the same net margin moves 0.96 pts just by changing the denominator**.
- Comparison table**: 27 cells, **√20 / △4 / ?3** — "not available" is **never written as 0**; each cell says why it cannot be trusted.

The single most valuable finding

- One check covers exactly one failure mode**. After swapping two columns, every in-table identity **still holds** (each period is internally consistent).
- What catches it is **cross-document reconciliation** — same date, same line item, two filings must agree: only **3 of 43** rows still matched.

Honest boundaries

- Sample is **one issuer × 8 periodic filings + eleven annual reports** ⇒ not extrapolated to industry statistics;
- 7 of 27 cells are currently unavailable** (parent-company table sharing a page; a bank's labels and column layout) — **flagging △ is a feature, not a defect**;
- The concept dictionary covers **36 concepts**, not a full chart of accounts; identities cover only the main lines of the three statements;
- Coverage is measured, not estimated**: on **layouts I had tuned against** concept hit rate is **29–34/36**; on **layouts I had never seen** it drops to **1–15/36** (3 of 11 issuers yield no table at all; **23/33** statements recovered) — **this is not a general-purpose filing extractor; it is reproducible extraction + checks over a given layout distribution**.

Entry: github.com/laa1991/fin-data-rulers (scripts + raw data + per-number recomputation) · full record [docs/一页纸.md](#) · lab log [docs/实验日志.md](#) (incl. the 14 silent failure modes I hit) · comparison table [data/coa/compare.md](#) (every cell with provenance)